

KALYANI GUGGILLA

Lead Data Engineer | Healthcare Data Migration, ETL & Interoperability

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PROFESSIONAL SUMMARY

Data engineer with 6+ years of production experience moving healthcare data between systems accurately and safely. Built and operated ETL/ELT pipelines, legacy-to-cloud migrations, and payer/provider data exchange for a HIPAA-governed managed care organization. Strong SQL and Python across Azure SQL, MS SQL Server, Oracle, and Teradata, with hands-on work in healthcare interoperability formats including HL7 v2, FHIR/REST APIs, EDI X12 (837/834/835), XML, and ICD-10 coded claims data. Known for source-to-target mapping, data validation and reconciliation, root-cause investigation of data defects, and documentation downstream teams actually use. Currently completing an M.S. in Artificial Intelligence (Data Analytics), hybrid program.

CORE COMPETENCIES

Data Migration & Imports: legacy system migration, data conversion, source-to-target mapping, business rules definition, cutover planning and rollback procedures, reconciliation, data export and validation

ETL / ELT Engineering: Azure Data Factory, Talend 7/8 and Talend Management Console, SSIS, batch and incremental loads, error handling, restartability, pipeline monitoring and alerting

Healthcare Data & Interoperability: HL7 v2, FHIR / REST APIs, EDI X12 (837, 834, 835), XML, ICD-10, claims, member, provider and clinical data, HIPAA-governed environments

Languages & Scripting: SQL, Python (scripting and automation utilities), R / RStudio, XML, JSON

Databases & Platforms: Azure SQL, MS SQL Server, Oracle, Teradata, Databricks, Microsoft Azure

Data Quality & Support: data quality frameworks, validation and reconciliation queries, defect triage, troubleshooting, root-cause analysis, Tier 2/3 production support, SLA management

Reporting & Analysis: Power BI, Tableau, KPI definition, Excel (advanced, pivot and statistical analysis)

Delivery & Collaboration: Agile / Scrum, Jira, GitHub, pull request review, CI/CD, technical documentation, cross-functional and remote collaboration, vendor coordination

AI-Assisted Development: Claude Code and Cursor in daily SQL/Python workflow with human review of every generated change; LLM fundamentals, prompt engineering, responsible AI

PROFESSIONAL EXPERIENCE

Associate Software Engineer, Data & Analytics

Commonwealth Care Alliance | Boston, MA | Nov 2021 - Dec 2025

- Designed, built, and operated ETL/ELT pipelines on Microsoft Azure using Azure Data Factory, Talend, and SSIS, extracting and transforming member, provider, claims, and HR data from legacy source systems into governed target datasets.
- Led an Azure SQL cloud migration from legacy on-premises databases and delivered it with zero business disruption to downstream systems, including source-to-target mapping, staged cutover, validation runs, and documented rollback procedures.
- Built EDI X12 and XML processing frameworks for compliant healthcare data exchange with internal teams, payers, providers, and external vendors within HIPAA-governed environments; now the enterprise standard.
- Engineered REST API integrations between Workday HRIS and downstream platforms, replacing manual file-drop processes with real-time synchronization and eliminating a recurring source of reporting error.
- Wrote and optimized complex SQL across Azure SQL, MS SQL Server, and Oracle to analyze, validate, and reconcile enterprise data, and reduced critical query execution time by 30% through indexing strategy and stored procedure refactoring.
- Built Python and SQL scripts and utilities for recurring data analysis, validation, and reconciliation tasks, reducing manual effort in load verification and defect troubleshooting.
- Established data quality checks and root-cause investigation practices, tracing defects through pipeline logs, execution plans, and API payloads, then coordinating remediation with the owning teams.
- Reviewed inbound and outbound data for gaps, defects, and transformation issues, documented findings, and drove resolution across engineering, analytics, and business partners.

- Provided Tier 2/3 production support: monitored job queues, triaged and logged incidents, performed root-cause analysis, and communicated status and resolution to business stakeholders under SLA.
- Authored and maintained process documentation, transformation mappings, and data workflow standards, creating a reference that reduced onboarding time and cut repeat questions from downstream teams.
- Reviewed peer code and data logic in GitHub pull requests and mentored analysts on SQL performance patterns, raising consistency and maintainability of team output.
- Used Claude Code and Cursor daily to accelerate SQL and Python development, reviewing every generated change for correctness, edge cases, and performance before merge.

Web Data Analyst

Quadrant Resources Pvt. Ltd. | Hyderabad, India | Apr 2020 - Dec 2020

- Evaluated and labeled web search-relevance training data supporting Microsoft Bing, applying detailed rubrics and resolving ambiguous edge cases to measurable quality standards.
- Performed data validation and quality checks on the UHRS platform, flagging systematic inconsistencies and contributing rubric clarifications adopted by the wider team.

Associate Analyst

Global Logics Pvt. Ltd. | Hyderabad, India | Aug 2018 - Mar 2020

- Contributed to machine learning data projects for Google Maps, Waymo, and USAA Insurance, annotating and validating datasets, flagging systematic labeling errors affecting downstream model accuracy, and partnering with QA to safeguard data integrity across delivery cycles.

SELECTED ACHIEVEMENTS

- Delivered an enterprise Azure SQL cloud migration with zero business disruption across all downstream systems.
- Reduced database query execution time by 30% through indexing and stored procedure optimization.
- Received the internal Talend Data Master Award and a performance bonus for high-impact migration delivery.
- Implemented EDI X12 and XML processing frameworks now standard across enterprise healthcare data exchange.

PROJECTS

Symbiotic World 3D | Sundai Club hackathon, 2026

- Multi-agent 3D world simulation with several interacting agent species; contributed the AI agent and LLM integration layer and the simulation data logic governing agent state and species interaction. sundai.club/projects/3223b796-fbc3-4c37-aa27-81a245f3d56e

Claims Processing Pipeline Root-Cause Investigation

- Diagnosed a 136% rise in claim processing time (6.2 to 14.6 hours average) over six months using SQL diagnostics; isolated a broken ICD-10 validation rule affecting Specialty Drug claims and specified a three-part fix projected to cut processing time 65%.

Concord AI - Compliance-First Generative AI Platform (Business Plan)

- Designed a governed LLM architecture for regulated financial services: RAG grounding on approved documents, a deterministic transaction layer separating what the assistant says from what it may do, per-decision audit logging, and human-in-the-loop escalation.

Bank Customer Attrition Analysis

- Analyzed 10,000+ customer records to quantify churn by age band, tenure, and transaction behavior; attrited customers averaged \$3,095 in transaction volume versus \$4,655 for retained, translated into targeted retention actions.

Full project portfolio: kalyanig-personal.github.io/Final-URL

EDUCATION & CERTIFICATIONS

M.S., Artificial Intelligence - Data Analytics Specialization

Indiana Wesleyan University, USA | Hybrid program | Class of 2027 (in progress)

Coursework: Foundations of Data Analytics, Data Visualization (R/RStudio), Machine Learning Fundamentals, Big Data, Data Mining, Text Mining, Large Language Models and Generative AI, Responsible AI, AI Capstone.

B.Tech., Electronics & Communications Engineering

JNTUH University, Hyderabad, India | 2014 - 2018

Certification: Talend for Data Integration and Big Data Professional - Edureka